



Examining the user experience of learning with a hologram tutor in the form of a 3D cartoon character

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Abstract

A hologram tutor is an instructional medium that has promising potential to serve as a substitute for actual teachers in classroom learning. Commonly, actual humans are used as the characters for the hologram tutor. Doubtlessly, this choice has its own advantages. However, in terms of development, it involves difficult and complicated steps. Utilizing 3D cartoon characters can be an alternative for developing a hologram tutor. However, a question arises about learners' acceptance and experience of a hologram tutor in 3D cartoon-like human form. Thus, the purpose of this research is to design and develop a 3D cartoon-like hologram tutor and examine aspects of the user experience (UX). The User Experience Questionnaire (UEQ) was used to acquire responses from students once they had completed the learning session with the hologram tutor. Particularly, the experiment was carried out on 30 university students ($N = 30$). From the results, it can be concluded that the UX level with a 3D cartoon-like hologram tutor was promising and positive. Therefore, this form kind of hologram tutor is a potential alternative for actual human characters in designing the hologram tutor medium.

Author Note I hereby notify that the manuscript has not been published, and is not under consideration for publication elsewhere.

The manuscript should be published in Education and Information Technologies as the article is a recommendation of a new approach in leveraging technology in teaching and learning. Where, this matter is closely related to the theme of the journal. Furthermore, not much educational technology research focuses on the design aspect of hologram in being a representative of a lecturer or teacher in the classroom.

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1 Introduction

With the rapid advancement in multimedia technology, the process of producing multimedia materials has become easier and easier (Zheng et al., 2019). Animation specifically has spread widely in multiple fields, including education (Chan, 2015; Kainz et al., 2013; Çakmak & Kühnapfel, 2000). Accordingly, innovation on animation's use in education needs to be maximized to ensure optimum effectiveness of teaching and learning activities (Shiu et al., 2020; Chiang & Tu, 2016; Chan, 2015). Hologram technology appears to be among the innovations with significant potential to be implemented in the education field (Kelion, 2018). Obviously, holograms have been widely used in various field, such as at airports to guide travelers (Jones, 2012), at live concerts to project actual human artists (Hye-Won & A-Ram, 2016) as well as animated character artists (Yan Lam, 2016), political speeches which can be beamed to multiple locations (Haddad, 2014), medical imaging (Haleem et al., 2020) and marketing (Tan, 2017).

Specifically in education, a hologram tutor can be used as a teacher representative in the classroom, and it has been observed to have positive effects on students' excitement as well as increasing their interest in learning (Wójcik, 2018). The use of a hologram tutor also allows students to feel as if they are in actual classroom conditions (Ghuloum, 2010). In particular, hologram tutor is a display of three-dimensional human or cartoon-like animated characters formed by the interference of light beams from coherent light source; for instance, LCD projectors. Usually, hologram tutor is presented on a actual human scale. Presently, educators have started to adopt this holographic technology for use in teaching and learning; however, this practice is still at an exploratory stage (Li & Lefevre, 2020).

The word hologram is composed of the Greek terms “holos,” for “whole view,” and “gram,” which means “written” (Ghuloum, 2010). The original concept of a hologram was the interaction of two light sources to produce a 3D visual (Geng, 2013). Additionally, holograms can be produced by projecting 2D visuals from a display device, such as a screen or projector, so that they appear in 3D form (Ahmad et al., 2015; Ghuloum, 2010). There are currently two methods of hologram display: a special transparent screen arranged in a pyramid shape for four-sided display as in

Fig. 1 Pyramid-shaped four-sided display hologram. Source: McLean (2015)

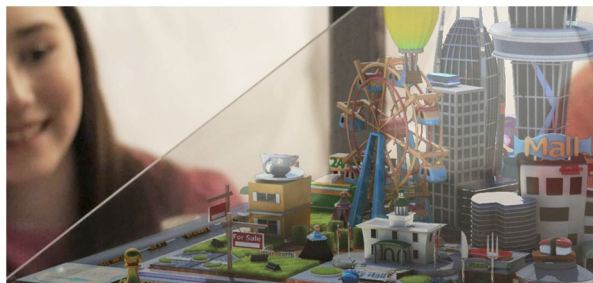


Fig. 2 Single-screen one-sided display hologram. Source: Binelli (2020)



Fig. 1 (Tiro et al., 2015; Reichelt et al., 2010) and a single screen for one-sided display as in Fig. 2 (Hong et al., 2014). Pyramid-shaped holograms are commonly used to visualize objects from four angles (Zeng et al., 2017). The advantage of four-sided display makes the pyramid-shaped hologram ideal for use as a medium in advertising and exhibition (Chaudhari et al., 2015). This is because the presentation can be viewed from various sides by groups of people. Although a pyramid-shaped hologram can be seen from various angles, it usually come in small and medium sizes. In comparison to pyramid-shaped holograms, holograms displayed on a single-screen for one-sided display can present visuals at full size or even larger (Leister et al., 2008). One-sided holograms are usually presented on stage for a focused audience (Kelion, 2018; Yang et al., 2016). Therefore, one-sided holograms are best used for character display, as they can present the character at the size of an actual human as if it were in front of the audience. Since this study is focused on character display, single-screen hologram was used.

Usually, realistic human beings are used as characters for hologram tutors. In fact, hologram tutors are said to be more attractive with realistic visualization (Yamada et al., 2018). This approach has been characterized as the right track for hologram development (Luévano et al., 2015). Accordingly, the process of upgrading hologram visualization has been carried out intensively, mainly in the production of visuals that resemble actual humans (Chang et al., 2018). The use of realistic human characters involves a specific studio development process, with a green- or black-screen background to ensure the hologram displays properly. This process is complicated and requires more development time.

Besides actual humans, characters such as cartoons, animals, robots etc. can be used in the hologram, accompanied by audio recorded from the actual teacher in the classroom. In this case, the design and development process only involves a computer. However, the question arises: What about user acceptance and UX with this approach? Therefore, the purpose of this study is to design and develop a hologram tutor in the form of a 3D cartoon character and analyze aspects of students' experience with it. For the purpose of this study, it was decided to use a less realistic character because such characters can promote positive emotions and increase the comfort level of the audience (Zibrek et al., 2018; Hamdan et al., 2015; MacDorman et al., 2009). However, previous studies only looked at the emotional effects of a 3D

cartoon character design displayed on screens, such as computer screens and smart-phones. This design's use as a hologram, which is displayed on a larger screen, has not been fully investigated yet.

2 Design and development

To ensure a systematic design and development process for instructional materials, it was imperative that the process be based on an appropriate instructional design model (Ali, 2018). There are numerous instructional design models, such as ADDIE, ASSURE, Dick and Carey, Waterfall and so on. Nevertheless, most of these models have roughly the same elements, namely analysis, design, implementation and evaluation (Ali, 2018). For the purpose of this study, the m-ADDIE instructional design model was selected. M-ADDIE is an acronym for five phases: analysis, design, development, implementation and evaluation; m stands for instructional multimedia. Each of the phases in m-ADDIE contains several subtasks that explain the function of each phase in a more detailed manner, specifically focusing on instructional multimedia materials (Ali, 2018).

2.1 Analysis

Analysis is the first task of the hologram tutor development process. It consists of several subtasks, which are deciding the method, selecting content, determining the target user, setting specific objectives and identifying appropriate software and platform (Ali, 2018).

The hologram tutor developed in this study focused on delivering content related to ethics for various age groups, from primary to tertiary education. Ethics was chosen because it is a subject that is easily understood by all students, ensuring that factors related to content difficulty do not affect students' acceptance and experience outcome. Difficult and complicated content might potentially increase students' cognitive load that most probably will influence their condition in learning (Feldon et al., 2019; Sweller et al., 1998). Therefore, by minimizing the cognitive load would possibly maximize the UX factors (Orozco, 2020; Margot, 2019; Zhou et al., 2011). Thus, easy to understand content was chosen to minimize the affect of this extraneous variable on UX outcome.

The specific topic selected was social media ethics, which concerns the benefits and drawbacks of using social media. In addition, the content of this topic was conveyed in a relaxed manner so as not to interfere with students' emotional responses in the classroom. Accordingly, the specific objective of developing this hologram tutor was that, after listening to the 30-min lecture by the hologram tutor, students would clearly understand the benefits and drawbacks of social media usage. The hologram tutor had no intelligent interaction features. However, students were able to request that the presentation to be repeated in whole or in part for better

understanding. As the study itself did not focus on comprehension, interaction was not an important variable.

To create the 3D cartoon character for use as this study's hologram tutor, Autodesk's 3ds Max software was used for the character design and animation process. The Adobe Premiere Pro program was used for audio editing, as well as video editing and preparation. Audio was derived from recordings of actual teachers in the classroom. The platform used was a digital video player in a computer for display through a high-resolution LCD projector.

2.2 Design

The design phase is the process of translating information from the analysis phase into sketches that will be referenced in the development process (Ali, 2018). Figure 3 depicts the hologram tutor design and development process in detail. The design and development process consisted of two subtasks: **(a)** design of the hologram tutor display panel or screen and **(b)** the character design of the hologram tutor.

For the design of the hologram display screen, several initial sketches were produced before the final drawing was completed. The design was based on numerous referenced studies, specifically ones related to the development of single-screen, one-sided hologram displays (Conner, 2020; Luévano et al., 2019; Hong et al., 2014; Figueiredo et al., 2014). The key factors considered in the design of the display screen were portability, transparency and a high-resolution projector with short-throw functionality, an audio system and systematic electronic management, as depicted in Fig. 4.

The second subtask in the design phase was creating the 3D cartoon character, as well as animating the hologram. 3D cartoon design has demonstrated capability

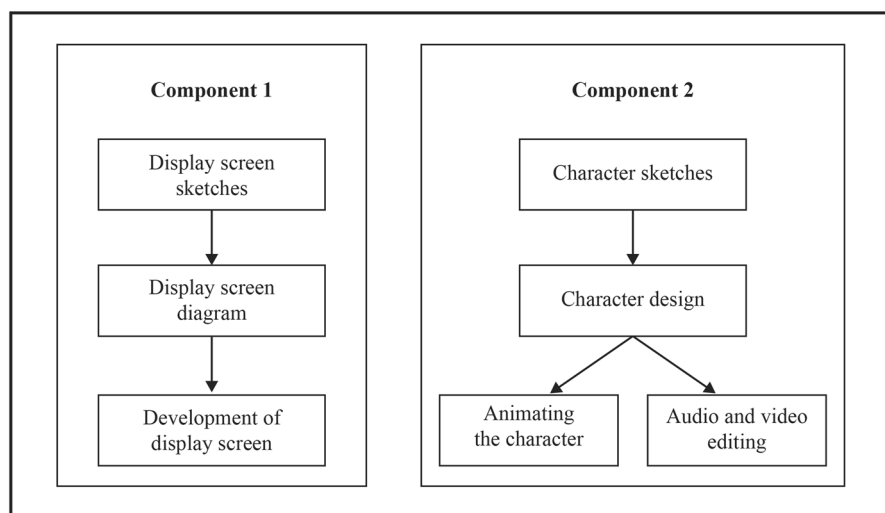


Fig. 3 Components of the hologram tutor design phase

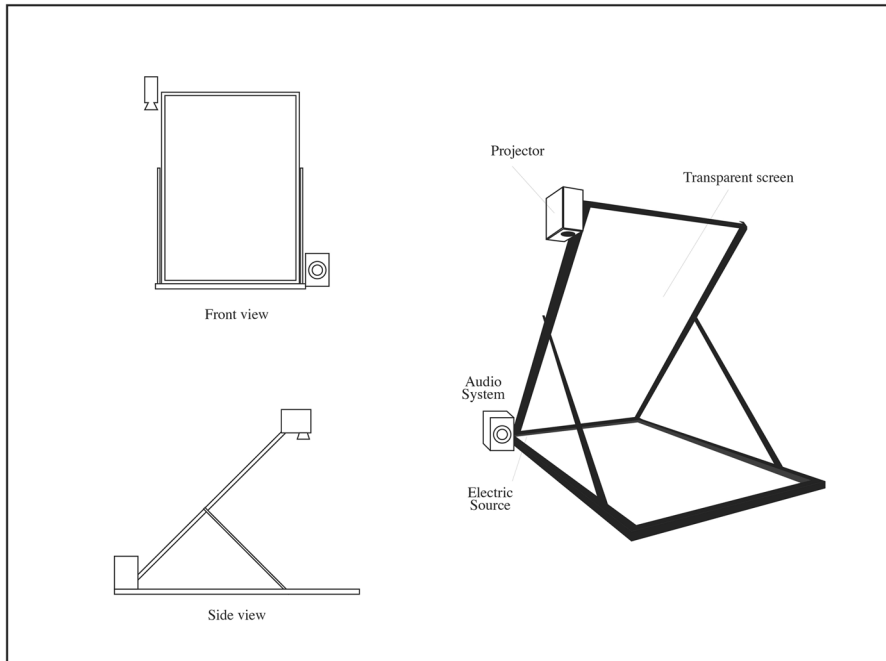


Fig. 4 Final drawing of the hologram display screen

to stimulate a sense of comfort and promote positive emotions among the audience (Hamdan & Ali, 2015; MacDorman et al., 2009). This argument was further supported by the uncanny-valley phenomenon, initially outlined by Mori (2012), whereby the level of realism in a character design might influence the comfort and emotion level of the audience. Many studies based on this phenomenon have been conducted in the fields of digital games and animation, and most found that 3D cartoon characters that are not too realistic or human-like can enhance the user's comfort level (Limano, 2019; Zibrek et al., 2018; Hamdan et al., 2015; Tinwell, 2015; McDonnell et al., 2012). With these findings in mind, initial sketches were made based on existing character models from the related websites. The character shown in Fig. 5 was finalized after going through an evaluation process involving experts and students.

Afterward, the animation for the hologram tutor was based on the recording of the actual teacher's lecture. The teacher's recorded movement was used as a guide throughout the animation process, such as in determining the body and lip movement of the hologram.

2.3 Development

The following phase is the development phase, which involves translating the specifications from the analysis and design phase into the final product. As described in

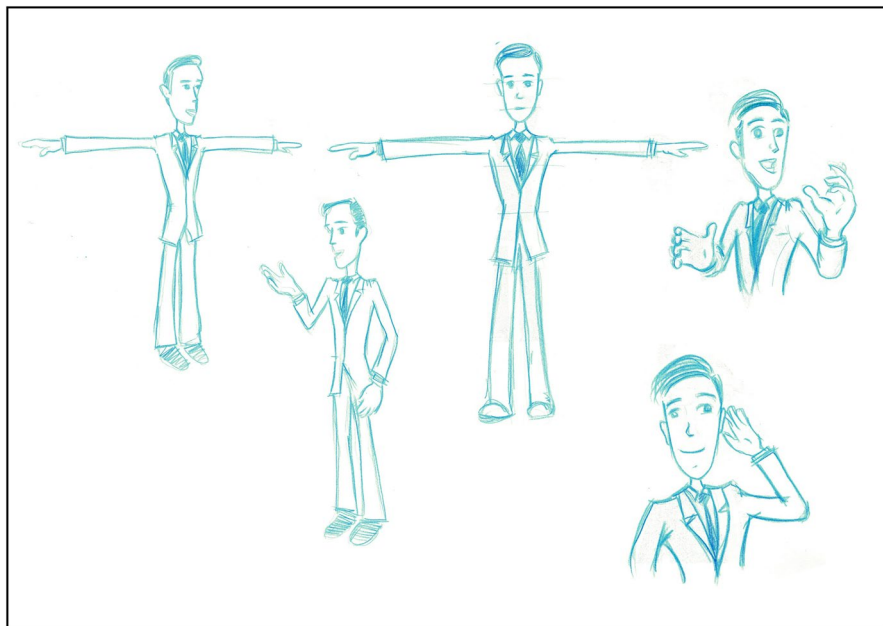


Fig. 5 Final sketch of 3D cartoon character

the previous section, two subcomponents of the hologram tutor were developed: the design of the hologram tutor display screen and character, as well as the hologram tutor animation.

The development of the display screen began with determining the appropriate structure for the display and support system. The projector requires a 45° forward-leaning, transparent display screen for the reflected image to appear at a right angle. The hologram tutor platform measures 5 ft wide and 10 ft high to obtain a display of 4 ft wide and 7 ft high, accommodating the average human size. In terms of support systems, the display platform requires the appropriate speaker, projector and electrical plug connection to be installed. Figure 6 depicts the final design of the hologram display screen.

Subsequently, the 3D cartoon character and its animations were developed. This process involves three main tasks: modeling, animation and final video editing. For the modeling process, a 3D cartoon character was constructed based on the final drawing from the design phase. Since the character modeling process requires specialized skills, an existing character was downloaded from the relevant website, as explained in the design phase. The character was subsequently modified to meet the requirements outlined in the design-phase sketches. Furthermore, the character's clothing was brightly colored to avoid interruption during the hologram display (Luévano et al., 2019; O'Connell & Rock, 2012).

In addition to the above, character development involved selecting appropriate skeletons and rigging to facilitate the animation process later. The animation was executed based on the recorded audio of the spoken script; audio needs to be



Fig. 6 Final design of the hologram display screen

recorded in advance to facilitate animation of features such as lip movements, facial reactions and body movements. The 3D animations were produced at high resolution for quality output (video mode 1080i, frame size 1920×1080 , 30fps). The 3ds Max software was used for modeling and animation tasks, while, in the final video editing process, Adobe Premiere Pro was used to add text and video, update color and edit audio (Figures 7, 8 and 9).

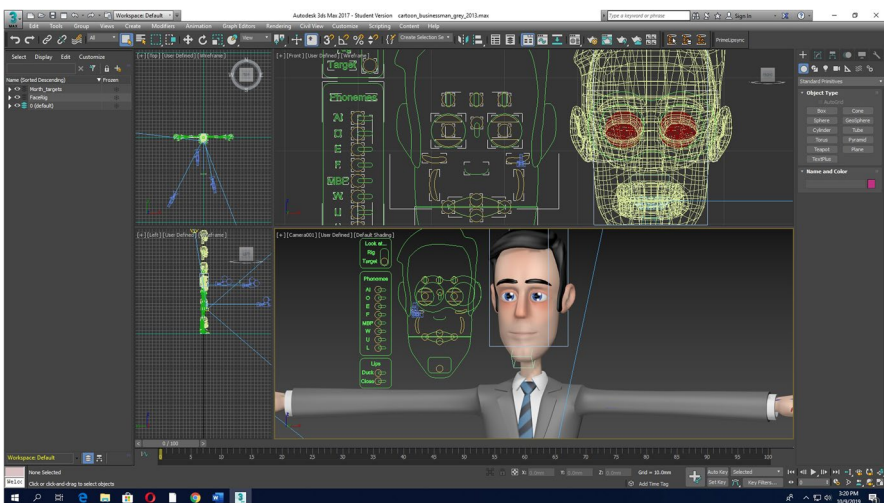
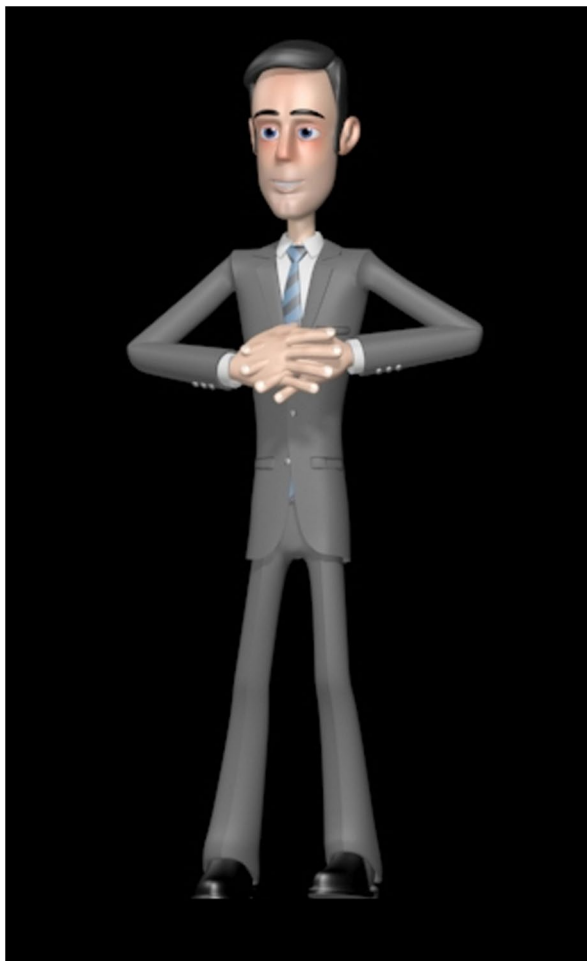


Fig. 7 Hologram tutor character modeling and animation using 3ds Max software

Fig. 8 Final design of 3D cartoon hologram tutor



2.4 Implementation and evaluation

Two methods of evaluation were carried out throughout the design and development process, namely formative and summative evaluation. The formative evaluation was performed during the process of hologram tutor design and development, and it involved assessment by both experts and students. Two animation experts, two content experts, an instructional technology expert and several students participated in the assessment process. For the most part, the assessments took the form of unstructured interviews. Based on the feedback, continuous improvements were made. Once the hologram tutor was completed, a summative evaluation in the form of a usability questionnaire was conducted among 30 students. Overall, the hologram tutor functions adequately and is well accepted by the students.

Fig. 9 Teaching session with hologram tutor



3 Methodology

This study aims to analyze the aspects of UX in learning with a hologram tutor in the form of a 3D cartoon character. To accomplish this, a 30-min teaching session with a hologram tutor was designed and developed.

3.1 Research sample

The study sample consisted of 30 individuals (16 men and 14 women) aged between 20 and 25 years old who were second-year bachelor's degree students at the Faculty of Art and Design, Universiti Teknologi Mara, Malaysia. The participants had no prior experience receiving instruction from a hologram tutor.

3.2 Research instrument

The UEQ (Schrepp, 2019) was utilized to obtain UX feedback regarding the learning process with the hologram tutor. When it was developed in 2005, the UEQ was only in German and English (Schrepp et al., 2014). Now, the UEQ is available in multiple additional languages including Malay, which was the preferred language for this research.

The UEQ contains six scales with 26 items and takes approximately three to five minutes to complete (Santoso et al., 2016), meaning it can optimize the use of time in obtaining users' feedback about products and applications. The UEQ was designed to allow users to respond quickly about the attitudes, impressions and feelings they experienced in using a product or application. The scales provided in the UEQ cover a wide range of UX attributes such as *attractiveness*, *perspicuity*, *efficiency*, *dependability*, *stimulation* and *novelty*. Each of the scales comprises four items, with the exception of attractiveness, which contains six. Attractiveness is considered a pure valence dimension. Meanwhile, perspicuity, efficiency and dependability are pragmatic quality dimensions, and stimulation and novelty are hedonic quality dimensions.

The feedback for each item on the UEQ consists of seven-point scales, ranging from 1 to 3 for a positive response, 0 for neutral and –1 to –3 for negative. The items

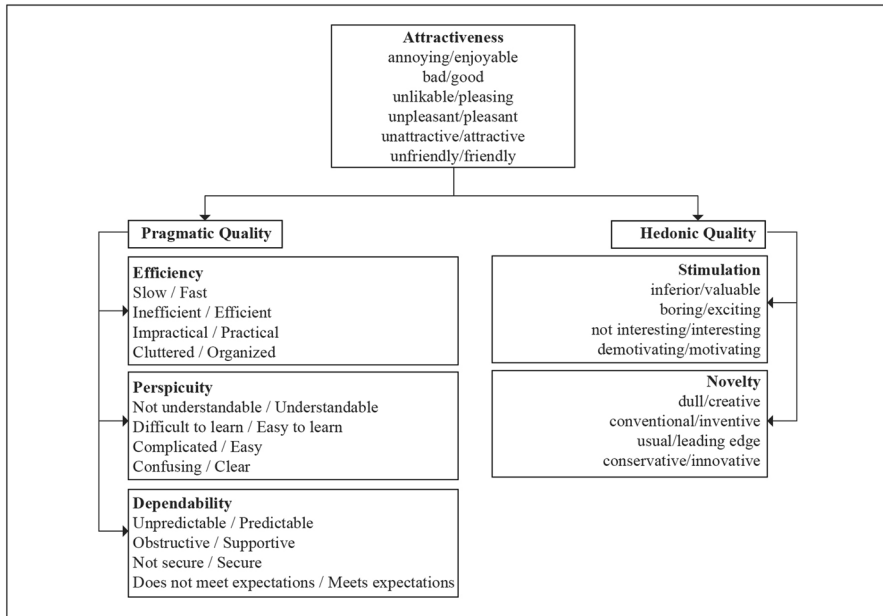


Fig. 10 Scales and items of the UEQ. Source: Schrepp (2019)

have a semantic differential form in that each item is represented by two opposite terms, for instance, “annoying” and “enjoyable,” which are used for an item in attractiveness. Figure 10 depicts the items in detail, as well as the scale structures of UEQ.

Findings from numerous previous studies have supported the UEQ as a reliable instrument in measuring UX, with high Cronbach’s alpha values (Schrepp, 2019; Kadastik et al., 2018; Schrepp et al., 2014; Laugwitz et al., 2008).

3.3 Research procedure

This study involved 30 students from the same class. They were required to attend a 30-min lecture session with the hologram tutor in a conducive classroom. The session was monitored by an independent lecturer who provided no additional learning information. No additional equipment, such as computers, smartphones and books, was used during the lecture session. The UEQ was distributed immediately after the students completed the hologram tutor session, with approximately five minutes allocated for students to complete the questionnaire.

Respondents were provided with detailed information regarding the procedure of addressing the questionnaire items. Respondents were also requested to respond spontaneously, being told that the first score that came to mind was the best selection for each item. Respondents were informed that there was no right or wrong answer. The confidentiality of the feedback was assured, and the feedback provided was only for the purpose

of this research. In addition, respondents were informed that they were free to answer the questionnaire and that it had nothing to do with their grade on the actual subject.

Once the questionnaires had been collected, checking was done to identify defective or incomplete questionnaire sets; none were found. Subsequently, the data obtained from the students were analyzed. Mainly, this study used descriptive quantitative analysis.

4 Analysis

The data obtained were quantitatively analyzed using SPSS software. The Cronbach's alpha value for the 26 UEQ items as a group is 0.72, which confirms high scale reliability. Next, the mean score and standard deviation were calculated for each of the UEQ items. A mean score of -0.8 to 0.8 is considered neutral, while values higher than 0.8 are positive and values lower than -0.8 are negative (Santoso et al., 2016). A detailed analysis was performed on the mean (M) and standard deviation (SD) values obtained (See Table 1).

Table 1 Descriptive analysis of the UEQ outcome

Item	M	SD	Left	Right	Scale
1	1.3	1.4	annoying	enjoyable	attractiveness
2	1.2	1.2	not understandable	understandable	perspicuity
3	1.5	1.7	creative	dull	novelty
4	1.1	1.9	easy to learn	difficult to learn	perspicuity
5	1.5	1.5	valuable	inferior	stimulation
6	1.2	1.7	boring	exciting	stimulation
7	1.8	1.4	not interesting	interesting	stimulation
8	0.8	1.7	unpredictable	predictable	dependability
9	-0.4	1.6	fast	slow	efficiency
10	1.0	1.8	inventive	conventional	novelty
11	1.7	1.7	obstructive	supportive	dependability
12	1.2	2.1	good	bad	attractiveness
13	1.3	1.5	complicated	easy	perspicuity
14	2.2	0.9	unlikable	pleasing	attractiveness
15	1.4	1.8	usual	leading edge	novelty
16	1.7	1.2	unpleasant	pleasant	attractiveness
17	0.8	1.7	secure	not secure	dependability
18	1.2	1.9	motivating	demotivating	stimulation
19	0.5	1.6	meets expectations	does not meet expectations	dependability
20	1.9	1.0	inefficient	efficient	efficiency
21	0.9	1.8	clear	confusing	perspicuity
22	1.1	1.8	impractical	practical	efficiency
23	1.2	1.6	organized	cluttered	efficiency
24	1.6	1.9	attractive	unattractive	attractiveness
25	0.8	1.6	friendly	unfriendly	attractiveness
26	1.6	1.8	conservative	innovative	novelty

4.1 Attractiveness

Attractiveness is an overall impression of the learning process with the product (Schrepp, 2019). As whole, students seemed to enjoy the learning process with the hologram tutor in 3D cartoon-character form. They agreed that they liked the hologram tutor ($M=2.2$, $SD=0.9$) and that the learning experience was pleasing ($M=1.7$, $SD=1.2$). The students also agreed that the hologram tutor was attractive ($M=1.6$, $SD=1.9$), enjoyable ($M=1.3$, $SD=1.4$), good ($M=1.2$, $SD=2.1$) and somewhat friendly ($M=0.8$, $SD=1.6$).

4.2 Pragmatic quality

Efficiency: This scale describes the UX in terms of the effort needed to complete a task (Schrepp, 2019). Overall, students agreed that they could complete the learning task without any unnecessary effort. Students agreed that the learning task with the hologram tutor was efficient ($M=1.9$, $SD=1.0$), practical ($M=1.1$, $SD=1.8$) and organized ($M=1.2$, $SD=1.6$). However, they seemed undecided on whether the learning process was fast or slow ($M=-0.4$, $SD=1.6$).

Perspicuity: This scale explains the ease of learning to use a product (Schrepp, 2019). Overall, students did not have problems in becoming familiar with the hologram tutor. They found that it was easy to learn ($M=1.1$, $SD=1.9$), understandable ($M=1.2$, $SD=1.2$) and not complicated ($M=1.3$, $SD=1.5$), since they only need to sit and listen. Overall, the learning process with the hologram tutor was clear for the students and did not confuse them ($M=0.9$, $SD=1.8$).

Dependability: This scale measures whether the users are in control of the interaction with the product (Schrepp, 2019). From the descriptive analysis, it can be seen that the hologram tutor appeared to be supportive and not obstructive ($M=1.7$, $SD=1.7$). However, the students were neutral when rating whether the hologram tutor felt predictable ($M=0.8$, $SD=1.7$) and secure ($M=0.8$, $SD=1.7$) and met their expectations ($M=0.5$, $SD=1.6$). These variables are closely related to features of smart interaction, which were not part of the hologram tutor.

4.3 Hedonic quality

Stimulation: This scale explains the UX in term of whether the product boosts excitement and motivation (Schrepp, 2019). Overall, it seems students were excited and motivated by the 3D cartoon hologram tutor. The students agreed that learning from the hologram tutor was interesting ($M=1.8$, $SD=1.4$), exciting ($M=1.2$, $SD=1.7$) and motivating ($M=1.2$, $SD=1.9$). They also acknowledged the value of the hologram tutor ($M=1.5$, $SD=1.5$).

Novelty: This scale concentrates more on the creativeness and innovativeness of the product (Schrepp, 2019). As whole, students agreed that that hologram tutor approach was creative ($M=1.5$, $SD=1.7$), innovative ($M=1.6$, $SD=1.8$) and leading edge ($M=1.4$, $SD=1.8$). The students also agreed that it was opposite conventional ($M=1.0$, $SD=1.8$).

In sum, the findings from the analysis indicate that all the subitems for attractiveness, pragmatic quality and hedonic quality were mostly in the positive range.

5 Discussion

Hologram tutors have a promising capacity to replace actual teachers in the classroom. Alternatively, holograms can be utilized as a medium to reduce the workload of a teacher or lecturer. With their science-fiction-like appearance and features, hologram tutors could also enhance students' interest and motivation in learning. To achieve this, the design of the hologram tutor requires particular attention. Typically, actual humans are used as the character for the hologram tutor. However, the use of actual humans is complicated and requires a relatively long period of time for development. The recording process calls for a specialized studio with the correct green- or black-screen background. Problems may also arise as the teacher or lecturer attempts to recreate the teaching process effectively in the recording studio.

To address this problem, appropriate 3D cartoon characters can be employed in conjunction with audio recorded from a teacher's actual classroom instruction. For this, the developers may only need a suitable workstation or computer system. However, questions arise about students' experience and acceptance of such an approach. The findings of this study address these questions in greater detail than previous research has. Apparently, overall, the students' experience and acceptance of the hologram tutor in the form of a 3D cartoon was mainly positive. In general, the UX test of the 3D cartoon hologram tutor with 30 tertiary-level students went smoothly. From the outcome of the data analysis conducted, it can be seen that the 3D cartoon hologram tutor attracts students' interest and increases their enjoyment of classroom learning. This finding was evidenced by the positive mean scores obtained for most attributes of attractiveness, pragmatic quality and hedonic quality.

In particular, attractiveness, which gives an overall view of the students' experience, indicates that the students agreed that the 3D cartoon hologram tutor is appealing and enjoyable. In addition to the hologram technology, the selection of a 3D cartoon character that is not too realistic might be among the factors that influence these findings. As Mori (2012) outlined, a less realistic character resembling an actual human has the potential to reduce user dissatisfaction. This finding is also in line with the findings of studies in the field of animation and digital games that examined the use of 3D cartoon characters that are not too realistic (Zibrek et al., 2018; Hamdan et al., 2015; MacDorman et al., 2009). However, further studies on the effects of 3D characters with various levels of realism may be needed to affirm this claim.

In addition, three aspects of pragmatic quality were investigated: efficiency, perspicuity and dependability. Efficiency explains whether the hologram tutor was easy to use and learn with. Overall, students feel that learning with the hologram tutor was efficient, practical and organized. However, the students expressed neutrality on whether the hologram's delivery of the course content was fast or slow. This was as had been expected, mainly based upon the observation done throughout the teaching process. Initially, students seemed excited, probably due to the new technology. However, their focus began to fade when the presentation took a long time to complete. It

can be concluded from this observation that the length of the lesson may affect the focus and concentration of the students. Therefore, it is important that further study is done to identify the appropriate presentation length for ideal learning with the hologram tutor. Further, there is a potential that students' attention spans to change when an actual human hologram is used. Study on this possibility is also crucial.

Perspicuity, the second aspect of pragmatic quality, explains whether students consider the 3D cartoon hologram tutor easy to use and learn with. Overall, the mean scores of each subitem of perspicuity were positive. This indicates that structured development based on an appropriate instructional design model has the potential ensure the quality and usability of a learning medium, as outlined in Ali (2018).

The third aspect of pragmatic quality is dependability, which explains whether hologram tutor users feel that they are in control of the interaction. Even though students agreed that the hologram tutor was supportive, they remained uncertain on the other aspects of dependability. This demonstrates that students were not confident that the use of the hologram tutor has potential to support and assist them in learning tasks. This might be due to the lack of bilateral interaction between the students and the hologram tutor, which left students as passive spectators. Additionally, the 3D cartoon character might cause students to presume that because the character is artificial, it is not as smart as an actual teacher. Therefore, further study on the components of interactivity and artificial intelligence needs to be done to address these issues.

Finally, this study examined the hedonic quality of the hologram tutor, focusing on two attributes: stimulation, which explains whether the hologram tutor is able to excite and motivate learners, and novelty, which explains whether the hologram tutor is an innovative and creative product that catches learners' interest. All the subitems of these two attributes were notably positive. These results align with findings that new technologies have potential in increasing students' interest and motivation (Sahin & Yilmaz, 2020; Wong et al., 2018; Papastergiou, 2009). Accordingly, efforts to implement 3D cartoon hologram tutors can be considered an alternative to having an actual teacher or lecturer in the classroom.

6 Conclusion

In conclusion, the study participants' feedback provides evidence that a hologram tutor in the form of a 3D cartoon character has a significant positive effect on students' interest and motivation. However, students were found to be less confident on the ability of the hologram tutor to assist them in their academic tasks. Students were also found to be unfocused if the length of the presentation is too long. Whether this was due to the design of the hologram tutor character remains a question. This might also be due to the fact that the hologram tutor is not an actual teacher. In light of this, students' confidence might be improved with the addition of interactivity and artificial-intelligence features. Further studies on this matter are necessary to find the best approach to using a 3D cartoon hologram tutor as a representative of the teacher or lecturer in the classroom.

This study was only limited in identifying the UX of utilizing 3D cartoon characters as alternative for developing a hologram tutor. Therefore, threats related to difficulty of the content and duration of the teaching session was not addressed

extensively. Future studies on courses with different levels of content difficulty, as well as various practical duration of teaching session is needed to further affirm the conclusion derived from this study.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request, depending on the approval of the university and the fund provider.

Declarations

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional research committee. Specifically, this study was approved by the Human Research Ethics Committee, Sultan Idris Education University, approval number 2019-0024-01 dated 19 June 2019.

Informed consent Informed consent was obtained from the institution involved, and all individuals included in the study participated voluntarily.

Competing interests The authors declare that they have no competing interest. The data that support the findings of this study are available from the corresponding author upon reasonable request, depending on the approval of the university and fund provider.

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